1 Dehydration of alcohols

$$\underset{\text{ethanol}}{C_2H_5OH} + \underset{\text{ethanol}}{C_2H_5OH} \xrightarrow[140\,^\circ C]{H_2SO_4} \underset{\text{ethoxyethane}}{C_2H_5OC_2H_5} + H_2O$$

2 Reaction with hydrogen iodide

$$\underset{\text{methoxymethane}}{CH_3{-}O{-}CH_3} + HI \xrightarrow{\Delta} \underset{\text{methyl iodide}}{CH_3I} + \underset{\text{methanol}}{CH_3OH}$$

$$\underset{\text{ethoxyethane}}{C_2H_5{-}O{-}C_2H_5} + HI \xrightarrow{\Delta} \underset{\text{ethyl iodide}}{C_2H_5I} + \underset{\text{ethanol}}{C_2H_5OH}$$

$$\underset{\text{methoxyethane}}{CH_3{-}O{-}C_2H_5} + HI \xrightarrow{\Delta} \underset{\text{methyl iodide}}{CH_3I} + \underset{\text{ethanol}}{C_2H_5OH}$$

3 Oxidation of alcohol

$$\underset{\text{ethanol}}{CH_3CH_2OH} + [O] \xrightarrow[\Delta]{K_2Cr_2O_7,H^+} \underset{\text{ethanal}}{CH_3CHO} + H_2O$$

(primary alcohol)

$$\underset{\text{2-propanol}}{CH_3CHOHCH_3} + [O] \xrightarrow[\Delta]{K_2Cr_2O_7,H^+} \underset{\text{propanone}}{CH_3COCH_3} + H_2O$$

(secondary alcohol)

4 Hydration of alkynes

$$\underset{\text{ethyne}}{HC{\equiv}CH} + H_2O \xrightarrow[HgSO_4]{dill H_2SO_4,60\,^\circ C} \underset{\text{ethanal}}{CH_3CHO}$$

$$\underset{\text{propyne}}{CH_3C{\equiv}CH_3} + H_2O \xrightarrow[HgSO_4]{dill H_2SO_4,60\,^\circ C} \underset{\text{propanone}}{CH_3COCH_3}$$

5 Oxidation of Aldehyde

$$\underset{\text{ethanal}}{CH_3CHO} + [O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7} \underset{\text{ethanoic acid}}{CH_3COOH}$$

$$\underset{\text{propanal}}{CH_3CH_2CHO} + [O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7} \underset{\text{propanoic acid}}{CH_3CH_2COOH}$$

6 Reduction

$$\underset{\text{ethanal}}{CH_3CHO} + 2[H] \xrightarrow[\text{dill } H_2SO_4]{LiAlH_4\text{in dry ether}} \underset{\text{ethanol}}{CH_3CH_2OH}$$

$$\underset{\text{propanone}}{CH_3COCH_3} + 2[H] \xrightarrow[\text{dill } H_2SO_4]{LiAlH_4\text{in dry ether}} \underset{\text{2-propanol}}{CH_3CHOHCH_3}$$

7 Addition with HCN



ethanal — 2-hydroxylpropanenitrile



propanone — 2-hydroxyl-2-methylpropanenitrile

8 Oxidation by acidified potassium dichromate

$$\underset{\text{ethanol}}{CH_3CH_2OH} + 2[O] \xrightarrow[\Delta]{K_2Cr_2O_7,H^+} \underset{\text{ethanoic acid}}{CH_3COOH} + H_2O$$

9 Oxidation by atmospheric oxygen

$$\underset{\text{ethanol}}{CH_3CH_2OH} + O_2 \xrightarrow{\text{bacteria}} \underset{\text{ethanoic acid}}{CH_3COOH} + H_2O$$

10 Reaction

$$\underset{\text{ethanoic acid}}{2\ CH_3COOH} + Mg^{2+} \longrightarrow \underset{\text{magnesium ethanoate}}{(CH_3COO)_2Mg} + H_2$$

$$\underset{\text{ethanoic acid}}{2\ CH_3COOH} + Na_2CO_3 \longrightarrow \underset{\text{sodium ethanoate}}{CH_3COONa} + CO_2 + H_2O$$

$$\underset{\text{ethanoic acid}}{CH_3COOH} + NaOH \longrightarrow \underset{\text{sodium ethanoate}}{CH_3COONa} + H_2O$$

Reduction

$$\underset{\text{propanoic acid}}{CH_3CH_2COOH} + 4[H] \xrightarrow[\text{dill } H_2SO_4]{LiAlH_4\text{in dry ether}} \underset{\text{4-propanol}}{CH_3CH_2CH_2OH} + H_2O$$

11 Esterification reaction

$$\underset{\text{ethanoic acid}}{CH_3COOH} + \underset{\text{ethanol}}{C_2H_5OH} \underset{\Delta}{\overset{H_2SO_4}{\rightleftharpoons}} \underset{\text{ethyl ethanoate}}{CH_3COOC_2H_5} + H_2O$$

$$\underset{\text{ethyl ethanoate}}{CH_3COOC_2H_5} + H_2O \underset{\Delta}{\overset{H_2SO_4}{\rightleftharpoons}} \underset{\text{ethanoic acid}}{CH_3COOH} + \underset{\text{ethanol}}{CH_3CH_2OH}$$

$$\underset{\text{ethyl ethanoate}}{CH_3COOC_2H_5} + NaOH \longrightarrow \underset{\text{sodium ethanoate}}{CH_3COONa} + \underset{\text{ethanol}}{CH_3CH_2OH}$$

12 Transesterification

$$\underset{\text{methyl ethanoate}}{CH_3COOCH_3} + \underset{\text{ethanol}}{C_2H_5OH} \underset{\Delta}{\overset{H_2SO_4}{\rightleftharpoons}} \underset{\text{ethyl ethanoate}}{CH_3COOC_2H_5} + \underset{\text{methanol}}{CH_3OH}$$

13 Methods of preparation

$$\underset{\text{bromoethane}}{C_2H_5Br} + \underset{\text{(in ethanol)}}{NH_3} \xrightarrow{\Delta} \underset{\text{ethanamine}}{C_2H_5NH_2} + HBr$$

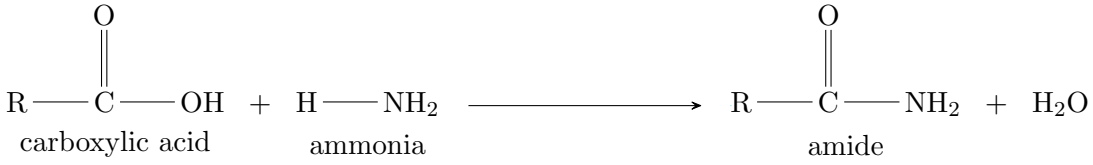
Note : ethyl bromide = bromoethane

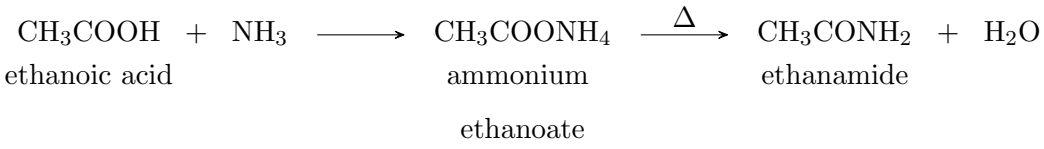
$$\underset{\text{ethanol}}{C_2H_5OH} + HBr \xrightarrow{H_2SO_4} \underset{\text{bromoethane}}{C_2H_5Br} + H_2O$$

14 Carbylamine reaction

$$\underset{\text{ethanamine}}{CH_3CH_2NH_2} + \underset{\text{(chloroform)}}{CHCl_3} + \underset{\text{(in alcohol)}}{3KOH} \longrightarrow \underset{\text{ethyl isocyanide}}{CH_3CH_2-N\equiv C} + 3KCl + 3H_2O$$

15 Condensation reaction





Hydrolysis

